BUSINESS PERFORMANCE REVIEW — VOLTEX ELECTRONICS (Q2–Q3 2024)

Guided by Maz and Lucas - *The Future Analyst*

Capstone project 2

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Today's Agenda

Key sections:

- Business Context & Performance Overview (Q2 → Q3 profitability shift)
- Profit Decomposition & Cost Drivers(Revenue vs COGS vs Return & OpEx)
- Return Cost Deep Dive(Region, Return Reason, Product Category)
- Operations Performance Validation(Delivery by Region & Shipping Partner)
- Customer Experience Impact(Rating & Sentiment linkage to returns)
- Key Insights & Recommendations(Short-term & Long-term actions)

EXECUTIVE SUMMARY

Profitability declined sharply in Q3 2024, with profit falling by ~47% while revenue declined by only ~16%, driven by elevated return-related and operational costs.

- Revenue declined noticeably in Q3 compared to Q2, while total costs showed limited reduction and higher volatility.
- As a result, profit and margins deteriorated more sharply than revenue, indicating pressure from cost-side inefficiencies.
- The negative impact was concentrated in specific regions and high-value product categories.
- Recovery should prioritize operational stabilization and targeted actions to reduce return and logistics-related costs.

BUSINESS PROBLEM & ANALYSIS SCOPE

Business Problem

Why did Voltex Electronics experience a profitability decline in Q3 2024 compared to Q2 2024, and which regions and product categories should be prioritized for recovery actions?

Business Objectives

- Assess the performance shift between Q2 - Q3 2024 with a focus on profitability.
- Identify key drivers behind the decline across revenue, cost, operations, and returns.
- Pinpoint affected regions and product categories to inform targeted recovery actions.

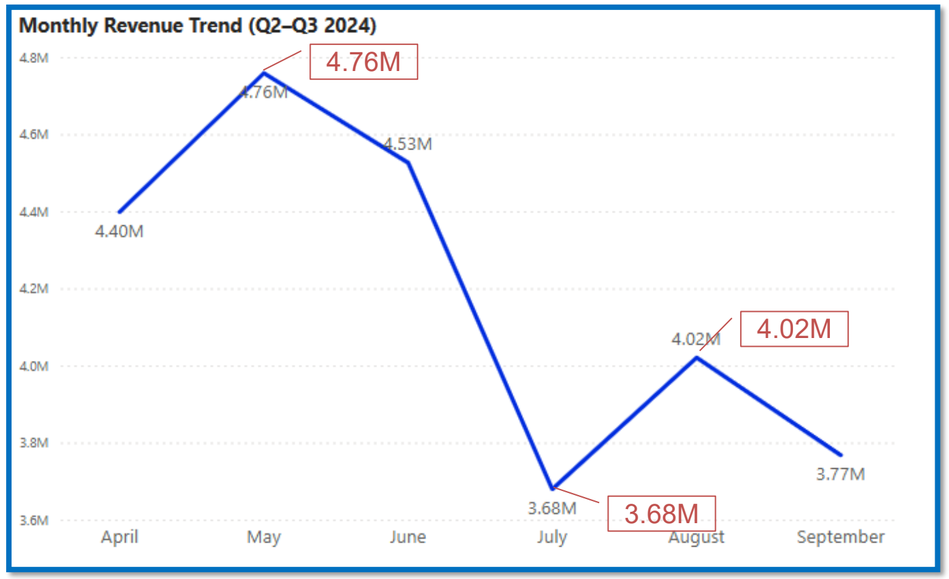
Analysis Scope

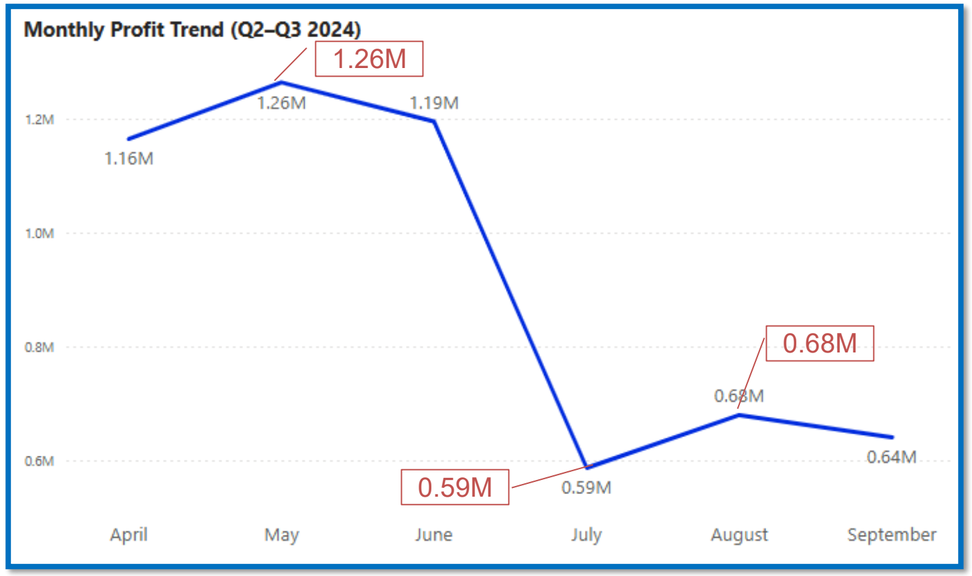
- In scope: Q2–Q3 2024 performance, profit and cost drivers, region and product category analysis, logistics and return-related factors, customer rating and sentiment.
- Out of scope: Long-term trends beyond Q3 2024, marketing or pricing strategy changes, external market factors.

Key Metrics (KPIs)

- Business Performance: Revenue, Orders, Profit
- Cost Drivers: Return Cost, Operational Cost
- Operations: Late Delivery %
- Quality & CX: Return Rate, Rating, Sentiment

Q3 performance deterioration was profit-led rather than revenue-led, signaling a structural margin issue rather than a demand shock.





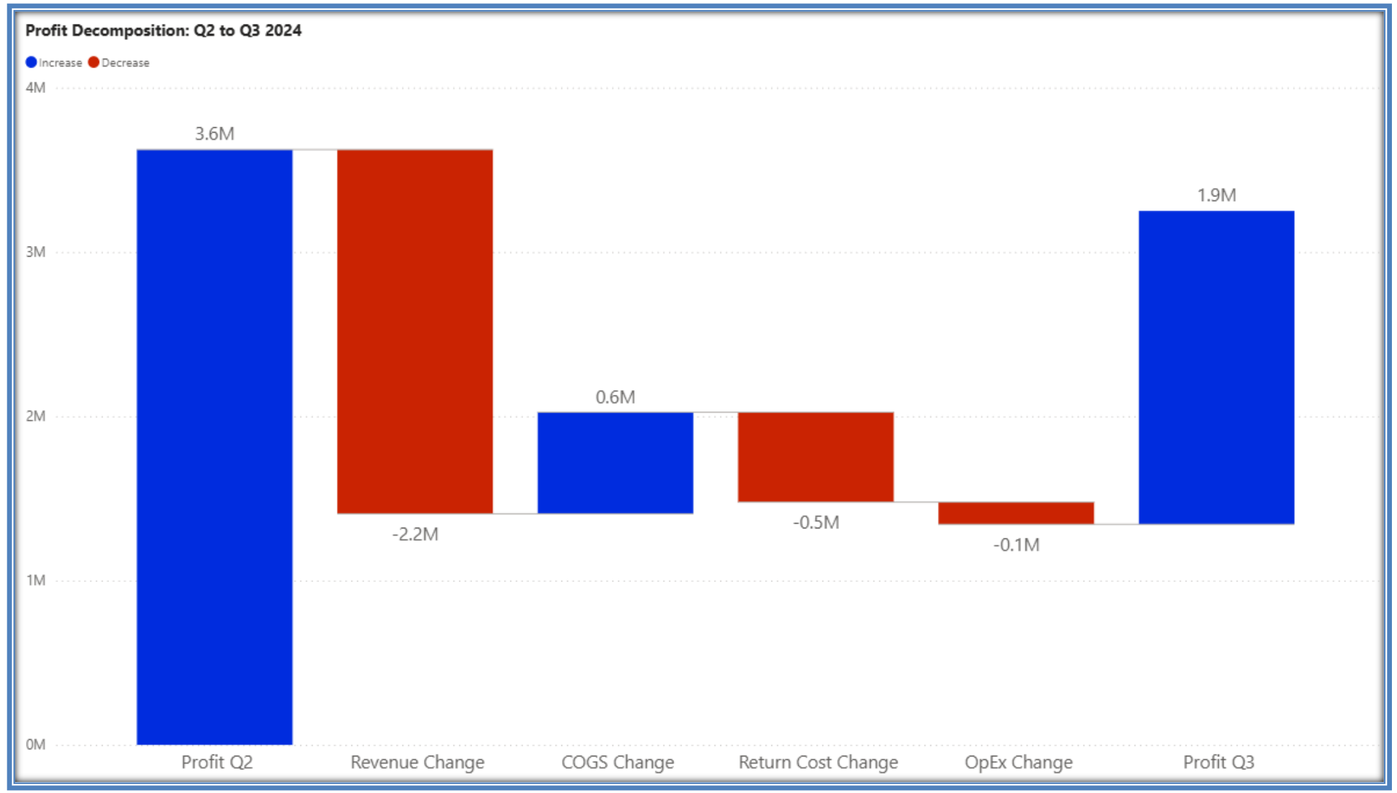
- While revenue declined moderately from a Q2 peak of ~$4.8M to ~$3.7–4.0M in Q3 (**≈15–20% decline**), **profit dropped disproportionately**, falling from >$1.2M to ~$0.6–0.7M (**≈45–50%).**
- The sharpest profit inflection occurred in **July**, where profit nearly halved month-over-month despite revenue remaining within Q3's normal range, indicating **cost pressure rather than volume collapse**.
- This divergence confirms that **Q3 underperformance is driven by margin erosion**, not topline volatility, warranting a focused investigation into **cost and operational drivers** rather than sales recovery alone.

Profit declined more sharply than revenue in Q3 as elevated return-related and operating costs amplified the impact of lower sales.

- **Profit declined by ~$1.7M from Q2 to Q3 (from ~$3.6M to ~$1.9M, ≈-47%),** significantly outpacing **the $2.2M revenue decline**, confirming that profitability deterioration was **not driven by revenue alone**.

- Despite **a $0.6M COGS improvement**, this benefit was more than offset by a **$0.5M increase in return-related costs** and **$0.1M increase in operating expenses**, indicating **limited cost flexibility under lower sales volume**.

- Overall, **return costs alone absorbed ~30% of the total profit decline**, demonstrating that **cost amplification—rather than demand weakness—was the critical driver of margin erosion in Q3**, warranting targeted investigation into return and operational drivers.

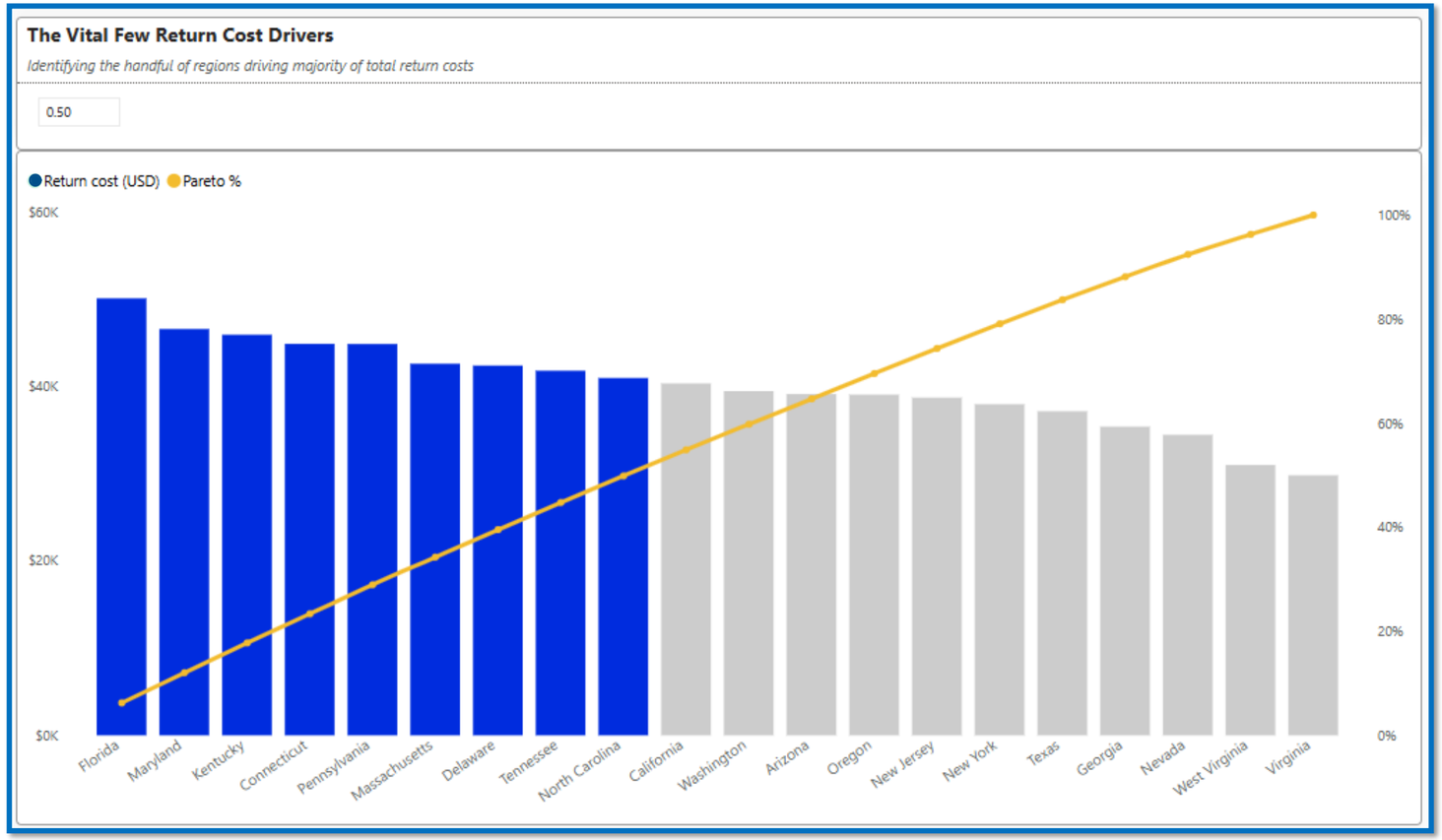


Return costs in Q3 are driven by a concentrated cluster of regions with consistently elevated return expenses, rather than isolated regional outliers—indicating a structural, multi-region issue.

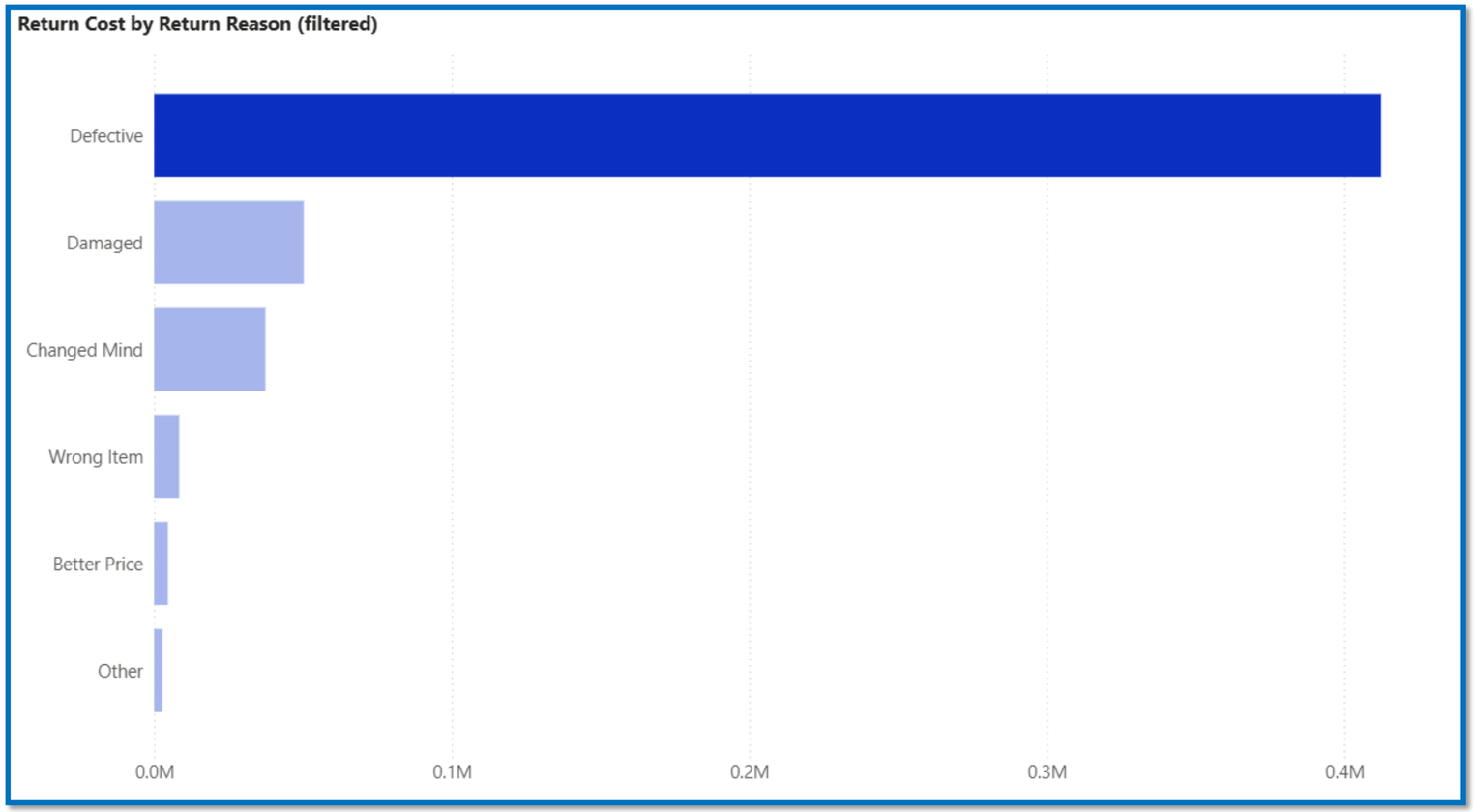
- **No single region dominates return costs in Q3**; instead, a group of regions with **similarly high return cost levels collectively accounts for the majority of total return expenses**, as shown by the gradual Pareto accumulation.

- **The absence of extreme outliers suggests the issue is systemic across a subset of regions**, pointing away from one-off regional failures and toward **shared underlying drivers** (e.g. product mix, quality exposure, or fulfillment patterns).

- **These regions therefore represent the highest-impact priority set for root-cause analysis**, warranting deeper investigation by **product category, return reason, and logistics execution** rather than region-specific fixes alone.

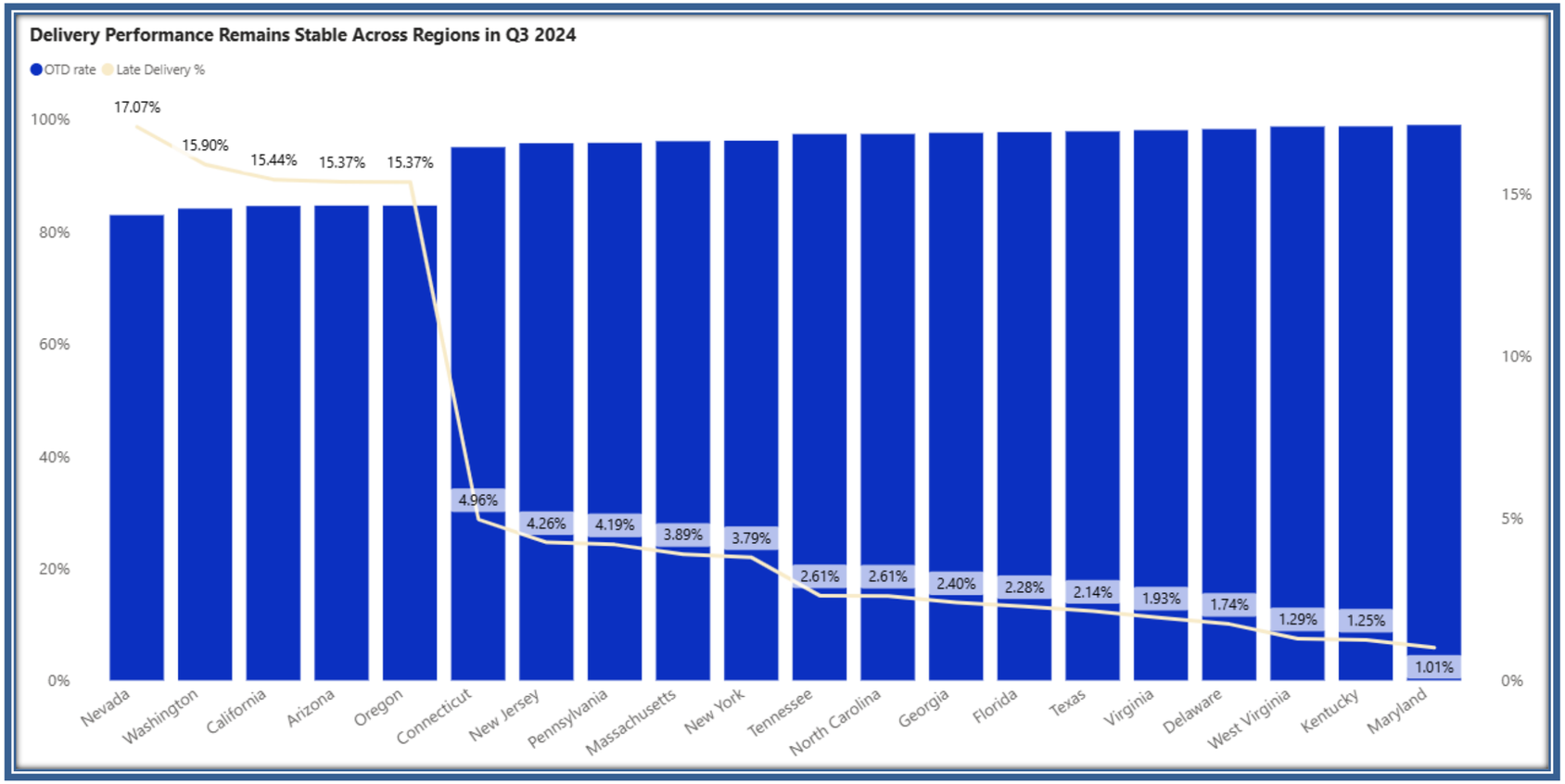


Return costs are elevated because defective products account for the majority of return expenses in priority regions



- **Defective products** alone contribute over **~$420K**, accounting for **~80%** of total return costs in the analyzed regions.
- **Non-quality-related** reasons (e.g. changed mind, pricing) collectively account for less than **~15%** of return costs, indicating limited behavioral or pricing-driven impact.
- This concentration confirms that elevated return costs are primarily driven by product quality issues, making quality control and defect prevention the highest-impact levers for cost reduction.

Delivery performance did not materially deteriorate in Q3; therefore, logistics delays are unlikely to be the primary driver of elevated return costs or profit decline.



- Most regions maintained strong on-time delivery performance in **Q3**, with OTD rates consistently **above 95%,** indicating stable logistics execution across the network.
- Late delivery rates remain low for the majority of regions, typically **below 5%**, suggesting that widespread delivery delays are not present.
- A small number of regions exhibit higher late delivery percentages (≈15–17%), but these are isolated cases and do not represent a systemic operational breakdown.
- The absence of broad logistics deterioration implies that elevated return costs are more likely driven by product quality issues rather than delivery performance.

Delivery performance issues in Q3 are driven by a single underperforming shipping partner rather than a systemic logistics breakdown.

192.28K
Return Cost – ReliableLog (Q3)

174.41K
Excess Return Cost – ReliableLog

- **ReliableLog's late delivery rate surged from 1.47% (Q2) to 15.82% (Q3)** — an increase of **+14.35 percentage points**, far exceeding historical baseline and peer performance (~2–3%).

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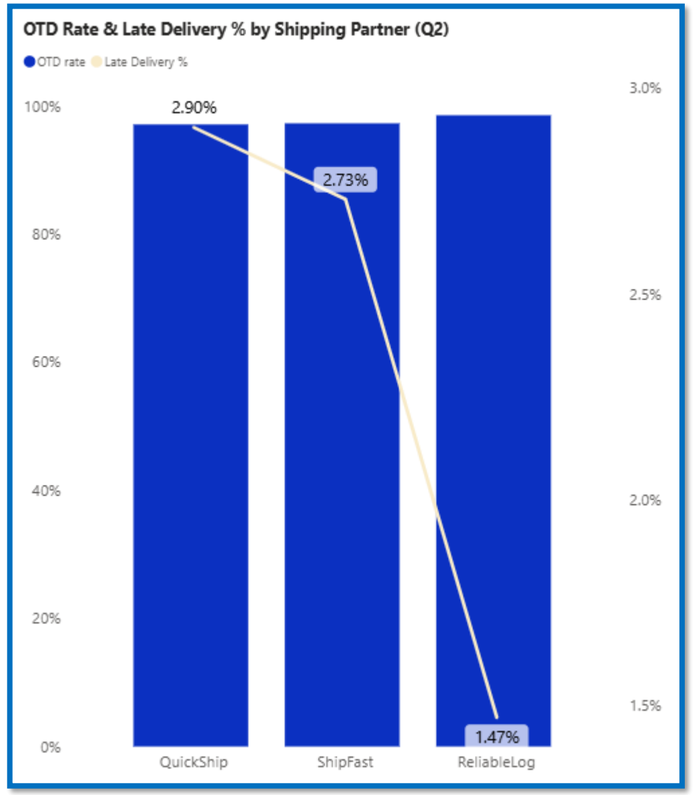
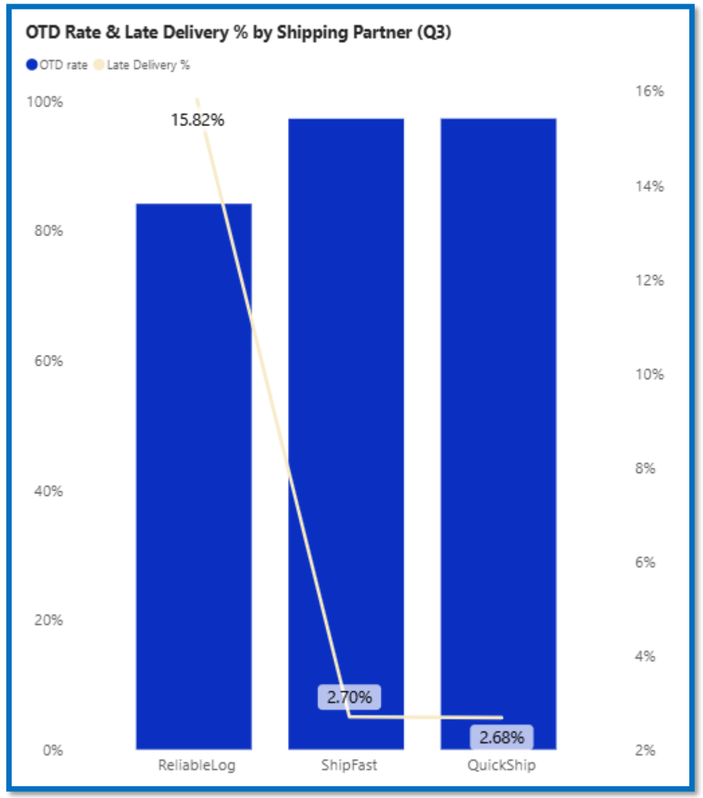
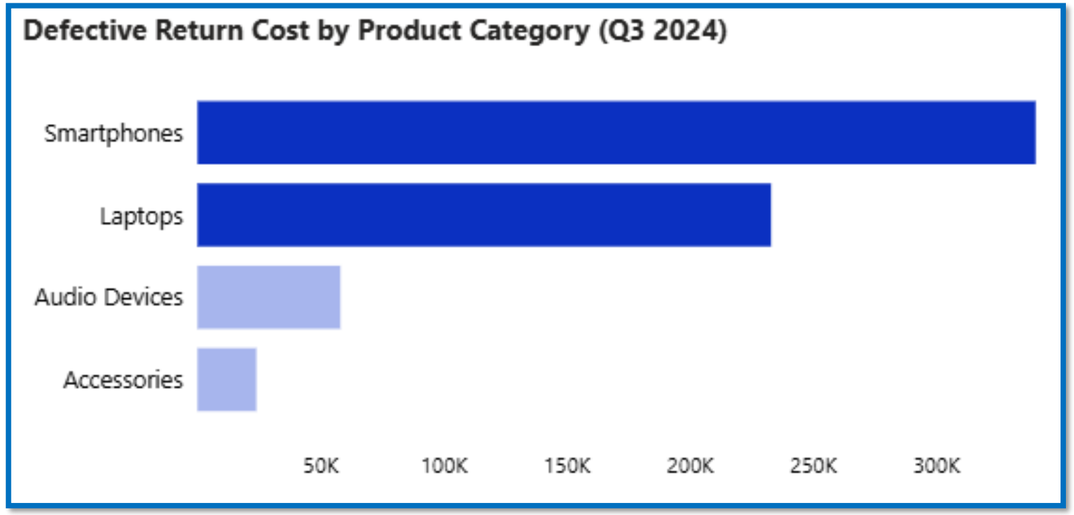
- This operational deterioration coincided with **$192K in total return costs in Q3**, of which **~$174K (~90%) represents excess return costs** attributable to abnormal late delivery rates.

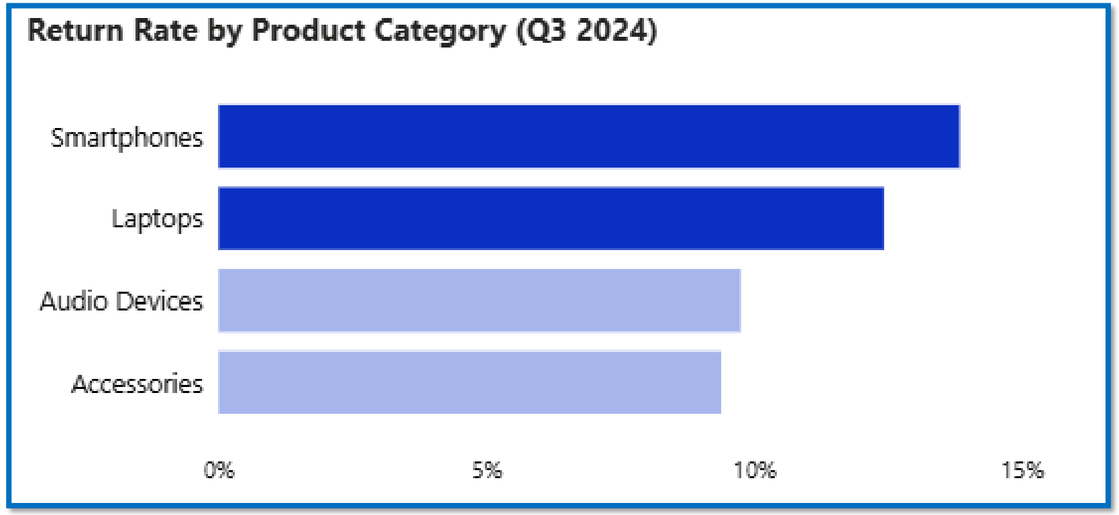
- In contrast, **ShipFast and QuickShip remained stable**, confirming that **return cost escalation is driven by a single partner failure rather than systemic logistics issues**.





Return costs remain elevated because Smartphones and Laptops consistently show both high return rates and high defective-related return costs in Q3.





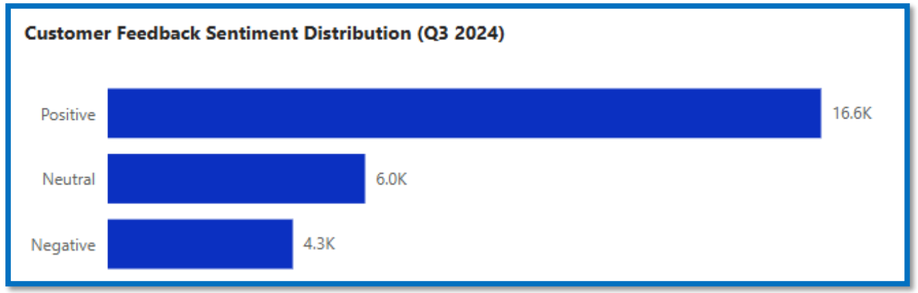
- Smartphones and Laptops account for the majority of defective-related return costs, contributing approximately **~$340K (Smartphones)** and **~$233K (Laptops)** in **Q3**, together representing **over ~85%** of total defective return costs, indicating a disproportionate financial impact compared to other categories.
- These same categories also exhibit the highest return rates, with **Smartphones at ~14%** and **Laptops at ~12%,** confirming that cost concentration is driven by elevated defect frequency rather than sales volume alone.
- Lower-cost categories (**Audio Devices, Accessories**) show materially lower **return rates (~9–10%)** and significantly smaller **return costs** (each below **~$60K**), suggesting limited contribution to overall return pressure.

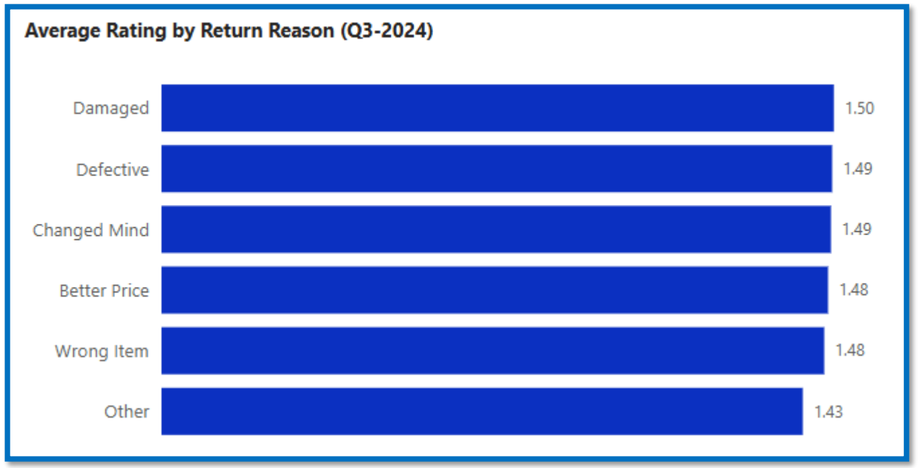
Customer satisfaction declined in Q3 because return-related issues—particularly defective and damaged products—are consistently associated with very low customer ratings and increased negative sentiment.

- **Return-related issues are strongly associated with poor customer experience** (Average ratings Q2 ~ 3.9, Q3 ~ 3.71), with all return reasons showing **very low average ratings of only ~1.4–1.5 out of 5** in Q3, indicating severe dissatisfaction once a return occurs.

- **Defective and damaged products—previously identified as major return cost drivers—also rank among the lowest-rated return reasons** (Defective: ~1.49, Damaged: ~1.50), directly linking quality issues to degraded customer experience.

- **Customer sentiment reflects this deterioration, with ~4.3K negative feedback entries (~16% of total feedback),** suggesting that return and defect incidents disproportionately drive negative perceptions despite a larger base of positive feedback.





RECOMMENDATIONS & NEXT STEPS

Short-term (0–3 months) — Stop the bleeding

• **Contain ReliableLog risk:** Address SLA breach and rebalance volume as ReliableLog's late rate jumped **1.47% → 15.82%**, driving ~**$174K excess return cost** in Q3.

• **Focus on high-impact categories:** Prioritize defect control for **Smartphones & Laptops**, which account for **>85% of defective return costs** and have the **highest return rates (~14% / ~12%)**.

• **Customer recovery actions:** Fast-track refunds and service recovery for defective/damaged returns to mitigate **very low ratings (~1.4–1.5/5)** and rising negative sentiment.

Long-term (3–12 months) — Fix the system

• **Partner performance governance:** Implement partner scorecards and early-warning thresholds to prevent partner-specific failures from escalating into major cost events.

• **Quality as a margin lever:** Embed defect and return cost KPIs into product and supplier decisions, targeting the **~80% of return costs driven by quality issues**.

• **Integrated CX monitoring:** Link **Defects** → **Returns** → **Delivery** → **Rating/Sentiment** in one dashboard to enable faster root-cause detection and cross-team accountability

Thank You

Questions & Discussion

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